

KENYA EBOLA SURVEILLANCE

Dashboard Validation Report and Developer Implementation Guide

Public-facing situation dashboard and restricted operational response workspace

IMPLEMENTATION STATUS

Validated direction for immediate BRD revision, UI/UX refinement, data mapping and development.

Date	3 August 2026
Version	1.0
Intended audience	Developers, UI/UX designers, data engineers, QA team, surveillance leads and the BRD team
Basis	Stakeholder review discussion and the public/operational dashboard screenshots shared during the session

Core outcome

The solution will have two dashboards only: a simplified public-facing situation dashboard and a restricted, role-based operational dashboard. The separate executive dashboard and the EOC Actions tab are removed from the target design.

1. Executive direction

This report converts the decisions made during the dashboard review into an implementable specification. Where the meeting reached a clear decision, the requirement is marked Approved or Remove. Where live or “gold” data is still required before a final decision, the item is marked Provisional or Validate.

ID	Requirement / decision	Status
D-01	Retain only two dashboard experiences: Public-facing and Restricted Operational.	APPROVED
D-02	Remove the separate Executive dashboard. Senior users who need more detail will use the role-based Operational dashboard.	REMOVE
D-03	Rebuild the public dashboard using the approved Figma prototype and use the existing page only as a content inventory.	APPROVED
D-04	Update the BRD immediately so it reflects the revised architecture, indicators, filters, system transition and access model.	APPROVED
D-05	Remove the EOC Actions tab from the operational navigation.	REMOVE
D-06	Use tab-specific filters, while keeping a consistent period selector across the operational dashboard.	APPROVED
D-07	Keep the first release simple. Do not expose indicators whose source, definition, coverage or quality cannot yet be defended.	APPROVED
D-08	Load and validate live/gold data before finalising unresolved POE, community and laboratory-detail logic.	VALIDATE

1.1 Immediate outcomes expected from the development team

- Revise the information architecture and routes so that only the public and restricted operational dashboards remain.
- Apply the approved Figma layout to the public dashboard without carrying forward duplicated text blocks or unnecessary notes.
- Reconfigure the operational tabs, filters, KPI cards, charts and detail tables according to Sections 5–10 of this guide.
- Prepare an indicator-to-source mapping and ensure every displayed value has a definition, calculation rule, source, refresh time and security classification.
- Update the BRD and use this document as the implementation decision log until the revised BRD is approved.

2. Purpose and scope

The dashboard must provide a clear national Ebola situation view to the public and a more detailed aggregate operational view to authorised response teams. The public view must communicate the situation quickly and plainly. The restricted view must support operational review without exposing patient or contact line lists.

The scope of this release covers dashboard architecture, navigation, filters, indicators, drill-down behaviour, data freshness, access control, source-system transition and the changes that must be captured in the Business Requirements Document (BRD).

3. Target dashboard architecture

Experience	Access	Purpose	Data level	Decision
Public-facing dashboard	Anonymous/public access	Concise national situation update, high-level aggregate metrics, public guidance and official help channels	National aggregate only; no operational records or line lists	Retain and rebuild in Figma design
Restricted operational dashboard	Role-based access for NPHI and specifically authorised senior/response users	Operational monitoring by tab, period, place and validated data source	Aggregate operational data with controlled drill-down; no public exposure of identifiable records	Retain and reconfigure
Executive dashboard	Previously separate	Duplicate executive view	Redundant with role-based operational access	Remove

Architecture principle

There must not be a permanent three-dashboard model. Senior leadership requirements are served through appropriate roles and permissions in the Operational dashboard, not through a separate Executive dashboard.

4. Cross-cutting design and functional requirements

ID	Area	Requirement
G-01	Data freshness	Show “Last updated” date and time prominently on both dashboards. Refreshing the page must not hide the age of the data.
G-02	Period selector	Replace “Last 14 days” as the main preset with Last 24 hours, Last 7 days, Last 21 days, Last 42 days and Custom date/time range.
G-03	Tab-specific filters	Only show filters relevant to the active tab. Do not carry County, Facility, POE, Result Status or other filters into tabs where they do not apply.
G-04	Drill-down	Operational KPI cards and charts should be selectable and open the exact aggregate records or breakdown used to calculate the value.
G-05	Reconciliation	Each summary number must reconcile to its detail table for the same period and filters.
G-06	Minimal wording	Use short, self-explanatory labels. Remove duplicated explanations, repeated notes and text that does not change the user’s decision.
G-07	Responsive layout	The dashboard must display correctly at common desktop widths without requiring browser zoom. Labels should wrap predictably and must not cause horizontal overflow.
G-08	Zero states	Zero values must be explicit. Do not display invented alerts, contacts, percentages or placeholder operational advice when the source value is zero or unavailable.
G-09	Source transparency	Show the source and coverage of each tab or indicator where this is needed to prevent users from assuming national completeness.
G-10	Production integrity	Prototype values must never appear as production data. Test or demonstration data must be visibly labelled and isolated from production.
G-11	Terminology	Write clinical/epidemiological terms in full where abbreviations could confuse users; for example, “Case Fatality Rate”.
G-12	Colour semantics	Use colour consistently, but do not rely on colour alone. Every value must have a clear text label.

5. Public-facing situation dashboard

The public dashboard will be rebuilt from the approved Figma prototype. The current public page is a source of existing content only; its layout and repeated sections should not be reproduced.

ID	Requirement / decision	Status
PUB-01	Use the approved Figma layout and visual hierarchy as the design source of truth.	APPROVED
PUB-02	Retain one clear “Current situation update” section with a visible data timestamp.	APPROVED
PUB-03	When the validated confirmed-case count is zero, display a concise dynamic statement such as “As of [timestamp], Kenya has no confirmed Ebola cases.” Do not hard-code the statement.	APPROVED
PUB-04	Remove duplicated daily brief text, repeated narrative and unnecessary free-text notes.	REMOVE
PUB-05	Do not provide public free-text operational notes. Any future note feature must be restricted, structured and separately approved.	REMOVE
PUB-06	Use a full-width/edge-to-edge situation image where indicated in Figma, while keeping the text area readable and uncluttered.	APPROVED
PUB-07	Keep public metrics high-level and plainly labelled. Explain units and avoid unexplained bars, percentages or contact-follow-up graphics.	APPROVED

ID	Requirement / decision	Status
PUB-08	Keep public action guidance and help channels concise and static unless an authorised content owner updates them.	APPROVED

5.1 Recommended public content order

1. National situation header and last-updated timestamp.
2. Concise current-status statement.
3. High-level metrics relevant to the public, with plain-language labels.
4. Essential public actions and official help channels.
5. Aggregate-data/privacy statement and official footer.

Public dashboard rule

The public page should answer three questions quickly: What is the current situation? What should the public do? Where can people obtain verified help? Anything that does not support these questions should be removed or moved to the restricted dashboard.

6. Restricted operational dashboard: final navigation

Tab	Decision	Purpose
Summary / Case status	Retain and redesign	National case cascade and facility-level breakdown
Laboratory	Retain and simplify	Tests done, positive tests and negative tests, with testing-laboratory breakdown
POE	Retain and simplify	Screening workload by point of entry and period
Health facilities	Retain / align with case-status table	Facility alerts, confirmed cases, current admissions, recoveries and deaths
Community	Retain provisionally	Signals reported and verified, subject to live-data and coverage validation
Contacts	Retain and simplify	Contacts associated with confirmed cases, filtered by place and period
EOC Actions	Remove	Not part of the target dashboard

6.1 Operational access model

- Require authentication and role-based access control.
- Grant operational access to NPHI and other specifically authorised senior/response users on the principle of least privilege.
- Do not expose patient or contact line lists in dashboard views. The dashboard should show aggregate operational data and controlled drill-downs only.
- Log access, data refreshes and administrative changes for audit purposes.

7. Summary and health-facility case status

The main operational summary should present a simple case cascade. The top cards must reconcile to the facility detail table. Only facilities with relevant data in the selected period should be displayed in the facility list or breakdown.

Order	KPI	Display rule
1	Alerts	Display the total number of alerts. Show suspected and probable cases as a clearly labelled secondary value or breakdown, not as an unexplained caption.
2	Confirmed cases	Display confirmed cases for the selected period; when the period is Last 24 hours, the value represents new confirmed cases in that period.
3	Current admitted	Display the number of currently admitted confirmed/probable cases according to the approved data definition.

Order	KPI	Display rule
4	Recoveries	Display cumulative or period recoveries consistently with the selected filter and clearly label the basis.
5	Deaths	Display deaths and a secondary "Case Fatality Rate" value. Handle a zero denominator without an error or misleading percentage.

7.1 Facility detail table

The facility table is the explanatory breakdown of the top cards. Its column totals must match the national KPI cards under identical filters.

Column	Rule
Facility	Facility with data/activity in the selected period
Alerts	Alert count attributable to the facility
Confirmed	Confirmed-case count
Current admitted	Currently admitted count
Recovered	Recovery count
Deaths	Death count

- Do not pre-populate the table with every facility in the country. Only show facilities with reportable activity for the selected period.
- The facility selector, where retained, should show facilities with relevant data rather than an unhelpful national list of inactive facilities.

8. Laboratory tab

The laboratory tab must be simplified to answer the core operational question: how many laboratory tests were completed, what were the results and where was the testing performed?

ID	Area	Requirement	Status
LAB-01	Top indicators	Use exactly three primary KPI cards: Tests Done, Positive Tests and Negative Tests.	APPROVED
LAB-02	Counting unit	The approved display unit is laboratory test events, not unique people. The label must say "tests" clearly.	APPROVED
LAB-03	Primary filters	Retain Period and Testing Laboratory. Remove the separate Facility dropdown and location/residence fields that confuse where the person came from with where the test was performed.	APPROVED
LAB-04	County/location	Do not use County of residence, sample-origin location or facility location as primary laboratory filters in this view.	REMOVE
LAB-05	Breakdown	Provide chart/table breakdown by testing laboratory. A card or chart selection should open the corresponding aggregate records.	APPROVED
LAB-06	Turnaround time	Do not make turnaround time a primary KPI in the first release. Add it later after its start/end definitions and data quality are validated.	DEFER
LAB-07	Pending/inconclusive	Do not silently classify pending or inconclusive results as negative. Final treatment in the KPI total requires data-team validation.	VALIDATE
LAB-08	Repeat tests	Do not deduplicate repeated samples into unique persons until person/sample linkage and the agreed epidemiological counting rule are validated.	VALIDATE

Laboratory total rule

The UI must not imply that Tests Done always equals Positive plus Negative until pending, inconclusive and repeat-test handling is agreed. The API and card definitions must explicitly state which result statuses are included.

8.1 Required laboratory drill-down fields

- Testing laboratory.
- Test event count.
- Result category.
- Selected period.
- Data source and last-updated timestamp.
- Where available, a non-public linkage key for quality review of repeated samples; this must not be displayed as identifiable patient data.

9. Point of Entry (POE) tab

The POE view must remain intentionally simple. The final discussion moved contact information into the Contacts tab, leaving screening as the primary POE indicator.

ID	Area	Requirement	Status
POE-01	Primary indicator	Number Screened.	APPROVED
POE-02	Filters	Period and POE only.	APPROVED
POE-03	Breakdown	Screening workload by POE and selected period.	APPROVED
POE-04	Origin and destination	Remove origin and destination filters/fields because the data may be self-declared, incomplete or not reliably linked to travel records.	REMOVE
POE-05	Investigation/alert statuses	Remove pending investigation, closed, escalated and similar status controls from the MVP until the source workflow is validated.	REMOVE
POE-06	Contacts	Do not duplicate contact indicators in POE; manage them in the Contacts tab with POE as a place filter.	APPROVED
POE-07	Detailed table	Populate the current POE detail table with live/gold data, review its behaviour and then decide which columns remain. Do not finalise placeholder columns.	PROVISIONAL

10. Community tab

Community surveillance data is currently limited by coverage, training, offline synchronisation, timeliness and validation concerns. The dashboard must not imply that the displayed values represent all community health promoters or national coverage.

ID	Area	Requirement	Status
COM-01	Provisional KPIs	Signals Reported and Signals Verified.	PROVISIONAL
COM-02	Remove duplicated contact metrics	Remove Contacts Traced from Community; contact indicators belong in the Contacts tab.	APPROVED
COM-03	Remove unlinked signals	Do not display "Unlinked Signals" as an MVP KPI.	REMOVE
COM-04	Coverage label	Display the actual source, participating counties/areas and data coverage. Do not label incomplete data as national.	APPROVED
COM-05	Definition of verified	Confirm whether "verified" means reviewed, verified true, or both true and false outcomes before finalising the metric.	VALIDATE
COM-06	Source role	Treat community HIS/eCHIS data as a secondary source until training, offline sync and timeliness are adequate for operational surveillance.	VALIDATE

11. Contacts tab

All contact-related metrics should be consolidated in one tab. This prevents users from seeing different contact numbers under POE, Community and Health Facilities without understanding the relationship.

ID	Area	Requirement	Status
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ID	Area	Requirement	Status
CON-01	Eligibility	Contacts are listed against a confirmed case. If there are no confirmed cases, contact counts and contact percentages must be zero.	APPROVED
CON-02	Primary filters	Period and Place/Source: All, POE, Health Facility or Community.	APPROVED
CON-03	Remove follow-up complexity	Remove follow-up channel, reached/missed, chatbot/WhatsApp, action and similar controls from the MVP.	REMOVE
CON-04	Remove actions section	Do not show a separate contacts “actions” area in the first release.	REMOVE
CON-05	Contact workflow	A contact who develops symptoms should move into the probable-case workflow, be tested and then become confirmed if positive; a negative result remains under the approved monitoring process.	APPROVED
CON-06	Twenty-one-day context	The system should support the 21-day monitoring period in the underlying workflow, but the first dashboard view should remain simple.	APPROVED

Zero-state safeguard

When confirmed cases = 0, the system must not show a non-zero contact count, a non-zero follow-up percentage or a fabricated contact workflow. API and UI tests must explicitly cover this condition.

12. EOC actions, notes and AI-generated advice

ID	Requirement / decision	Status
EOC-01	Remove the EOC Actions tab from the dashboard navigation.	REMOVE
EOC-02	Do not treat “operational checks” or AI-written recommendations as validated indicators.	APPROVED
EOC-03	Do not display a free-text note area on the public dashboard.	REMOVE
EOC-04	A future restricted notes/action-item feature may be considered only after ownership, workflow, expiry, approval and audit requirements are defined.	DEFER

13. Indicator data contract and API behaviour

Every metric should be implemented from an agreed data contract rather than from a UI label alone. At minimum, the backend should provide the following metadata with each indicator or make it available through an indicator registry.

Field	Requirement
indicator_id	Stable machine-readable identifier
display_name	Approved user-facing label
definition	What is counted and what is excluded
counting_unit	Cases, persons, contacts, tests, samples, signals or screenings
source_system	ADAM, UHAI, LIMS, laboratory register, community HIS/eCHIS or other approved source
coverage	National or named participating locations
refresh_frequency	Expected update cadence; current operational expectation is at least daily/within 24 hours unless automated more frequently
period_start / period_end	Exact time window applied
allowed_filters	Only filters that are valid for the indicator
breakdown	Rows used to reconcile the total
last_updated	Timestamp of the source data, not only the browser refresh time

Field	Requirement
security_classification	Public aggregate or restricted aggregate
data_quality_status	Validated, provisional, stale, incomplete or unavailable

13.1 API and UI safeguards

- Apply filters server-side and return the exact filter context with the response.
- Return a clear stale-data or unavailable-data state instead of zero when the source failed to refresh.
- Prevent division-by-zero errors for percentages and rates.
- Do not infer missing categories. Pending, unknown and inconclusive values must remain distinct until an approved rule exists.
- Expose the reconciliation breakdown used to calculate each card.
- Ensure card, chart and table values are derived from the same endpoint or shared calculation service.

14. Current systems and transition requirements

The BRD and system design must distinguish the current-state tools from the target state. ADAM and UHAI should not be presented as two permanent parallel systems. The target is one coherent surveillance solution with a managed transition and clear data hand-offs.

System/source	Current role	Implementation requirement
ADAM	Currently used at other points of entry and supports case investigation/follow-up functions.	Retain in the As-Is description and data-source mapping. Do not delete its references from the BRD without an approved transition decision.
UHAI	Currently used for screening at JKIA according to the discussion.	Document its current screening role and the hand-off to case investigation when a screened traveller becomes an alert/case.
LIMS / laboratory registers	Provide laboratory result data.	Map laboratory identifiers, result categories, timestamps and repeat-test linkage before finalising KPI calculations.
Community HIS/eCHIS	Provides limited community signals in selected areas.	Label coverage and treat as provisional/secondary until operational readiness improves.

14.1 Required transition design

- Describe one target system and a time-bound migration/transition approach.
- Define the screening-to-investigation hand-off so users do not enter the same event in two systems unnecessarily.
- Document which system is authoritative for each indicator during the transition.
- Provide deduplication/linkage rules across systems before combining records.
- Identify training, deployment, financing and connectivity dependencies that may affect the transition timeline.

15. Prioritised development backlog

Priority	Workstream	Deliverable
P0	Architecture	Remove Executive dashboard and EOC Actions route; retain Public and Operational only.
P0	BRD	Revise dashboard architecture, indicator tables, filters, access model and ADAM/UHAI transition.
P0	Public UI	Implement approved Figma layout, timestamp and dynamic no-confirmed-case state.
P0	Operational filters	Implement consistent period presets plus custom date/time and tab-specific filters.
P0	Core KPIs	Implement case-status, laboratory, POE and contacts rules that are already approved.
P0	RBAC/data safety	Enforce restricted access and prevent line-list/PII exposure.
P1	Reconciliation	Connect cards, charts and tables to the same calculation service and add drill-down.

Priority	Workstream	Deliverable
P1	Live data validation	Load gold data for laboratory, POE and community tabs and complete unresolved decisions.
P1	Coverage/freshness	Display source, coverage, last updated and data-quality state.
P2	Advanced laboratory metrics	Add turnaround time only after definition and quality approval.
P2	Contact follow-up detail	Add channels, missed/reached and follow-up analytics only when the workflow and data are reliable.
P2	Controlled notes/AI	Consider restricted notes or AI assistance only after governance and approval controls are defined.

16. Acceptance criteria for QA and sign-off

ID	Acceptance test
AC-01	Only Public and Restricted Operational dashboards are accessible; no separate Executive dashboard route remains.
AC-02	The EOC Actions tab is absent from navigation and direct URLs are disabled or redirected appropriately.
AC-03	The public page matches the approved Figma structure and contains no duplicated daily brief/note sections.
AC-04	Both dashboards display the source-data “Last updated” date and time.
AC-05	Operational period presets include 24 hours, 7 days, 21 days, 42 days and custom date/time.
AC-06	Each tab shows only its approved filters.
AC-07	Summary KPI totals equal the facility-table totals under identical filters.
AC-08	Laboratory top cards show Tests Done, Positive Tests and Negative Tests; Facility and residence/location filters are absent.
AC-09	Pending/inconclusive laboratory results are not silently converted to negative.
AC-10	POE shows Number Screened by POE and period; Origin, Destination and investigation-status controls are absent.
AC-11	Community coverage is labelled and does not imply nationwide reporting where coverage is limited.
AC-12	Contacts are centralised in the Contacts tab and can be filtered by POE, Health Facility or Community.
AC-13	When confirmed cases are zero, contacts and contact percentages are also zero and no fabricated workflow appears.
AC-14	Only facilities with relevant data are shown in facility breakdowns.
AC-15	Unauthorised users cannot access restricted routes or data through the UI or API.
AC-16	Prototype/test values are not mixed with production data.

17. Open decisions and dependencies

ID	Decision needed	Reason
O-01	Laboratory repeat tests	Confirm whether future reporting needs a separate “unique positive persons” measure in addition to test-event counts.
O-02	Laboratory result statuses	Agree how pending, inconclusive, invalid and repeat results affect Tests Done and the detail table.
O-03	POE detailed table	Review with live/gold data and approve the final columns.
O-04	Community verification	Define verified, verified true/false, source coverage and operational timeliness.
O-05	ADAM/UHAI transition	Approve the target platform, migration sequence, hand-off, training and timelines.

ID	Decision needed	Reason
O-06	Figma version	Confirm the exact approved Figma file/version and freeze the layout for implementation.
O-07	Rate zero-state	Confirm the display convention for Case Fatality Rate when confirmed cases are zero: N/A or 0%, while preventing calculation errors.
O-08	Refresh mechanism	Confirm whether the 24-hour update is batch, automated or manual upload and define the stale-data threshold.

18. BRD revision checklist

- Replace the three-dashboard model with the two-dashboard architecture.
- Remove the Executive dashboard and EOC Actions tab requirements.
- Insert the final navigation and indicator tables from this guide.
- Specify public versus restricted data classification and role-based access.
- Add the period presets, custom date/time selector and tab-specific filtering rule.
- Document indicator definitions, calculation units, source systems, refresh cadence and drill-down behaviour.
- Document current ADAM/UHAI use, the screening-to-investigation hand-off and the single-system target state.
- Record unresolved items as explicit assumptions/decisions required rather than allowing developers to guess.
- Add the QA acceptance criteria in Section 16 to the requirements traceability matrix.

Implementation instruction

Where this report marks an item Validate, Provisional or Deferred, developers should not invent the business rule. Build the component so the rule can be configured, load the validated live data and return the item to the product/surveillance team for sign-off.

Appendix A. Reference screens reviewed during the session

These images are included only to identify the current content and prototype areas discussed. They are not substitutes for the approved Figma design or the requirements in this report.

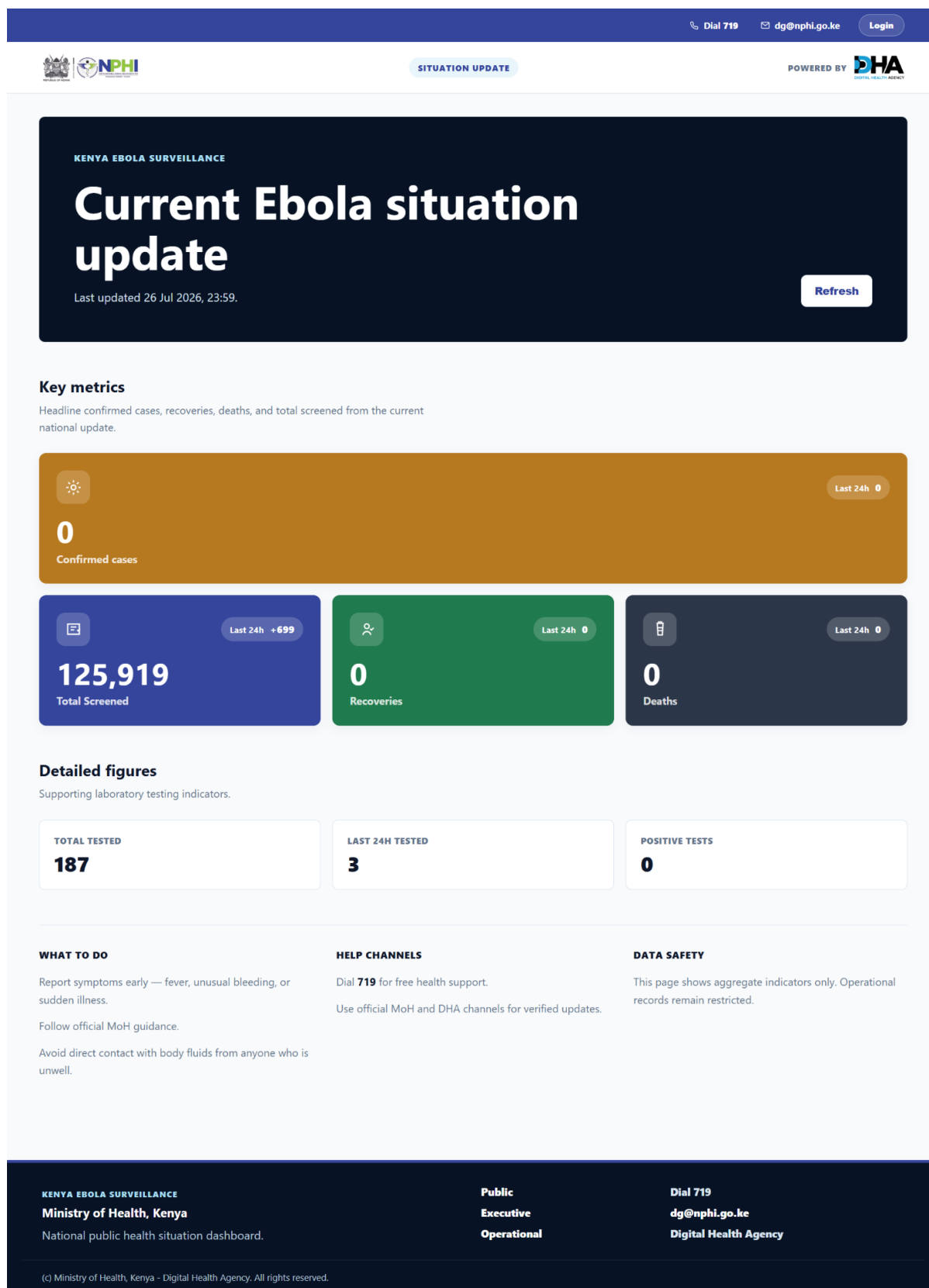
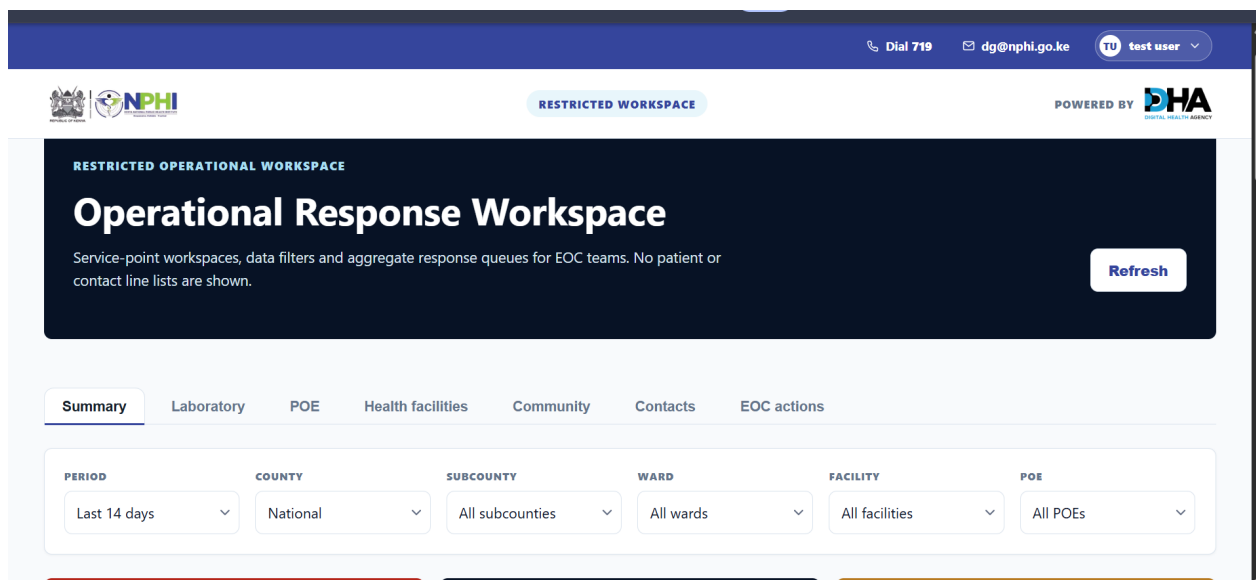


Figure A1. Current public-facing dashboard used as a content inventory; the layout is to be replaced by the approved Figma design.



RESTRICTED WORKSPACE

POWERED BY **DHA**

Operational Response Workspace

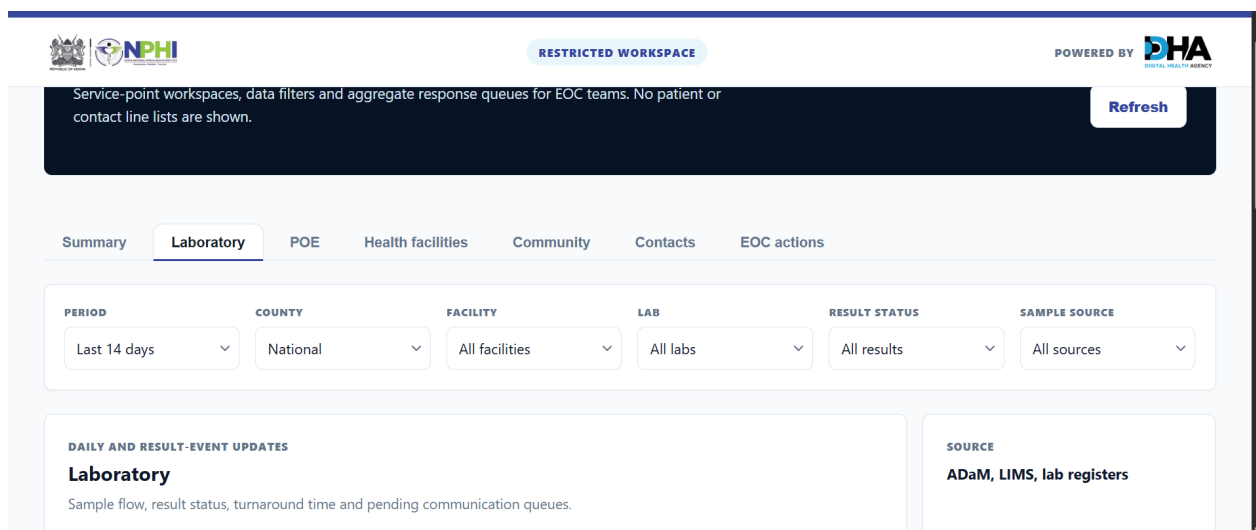
Service-point workspaces, data filters and aggregate response queues for EOC teams. No patient or contact line lists are shown.

[Refresh](#)

Summary | Laboratory | POE | Health facilities | Community | Contacts | EOC actions

PERIOD: Last 14 days | COUNTY: National | SUBCOUNTY: All subcounties | WARD: All wards | FACILITY: All facilities | POE: All POEs

Figure A2. Restricted operational workspace and current tab/navigation prototype.



RESTRICTED WORKSPACE

POWERED BY **DHA**

Laboratory

Service-point workspaces, data filters and aggregate response queues for EOC teams. No patient or contact line lists are shown.

[Refresh](#)

Summary | **Laboratory** | POE | Health facilities | Community | Contacts | EOC actions

PERIOD: Last 14 days | COUNTY: National | FACILITY: All facilities | LAB: All labs | RESULT STATUS: All results | SAMPLE SOURCE: All sources

DAILY AND RESULT-EVENT UPDATES

Laboratory

Sample flow, result status, turnaround time and pending communication queues.

SOURCE

ADaM, LIMS, lab registers

Figure A3. Laboratory filters before simplification.

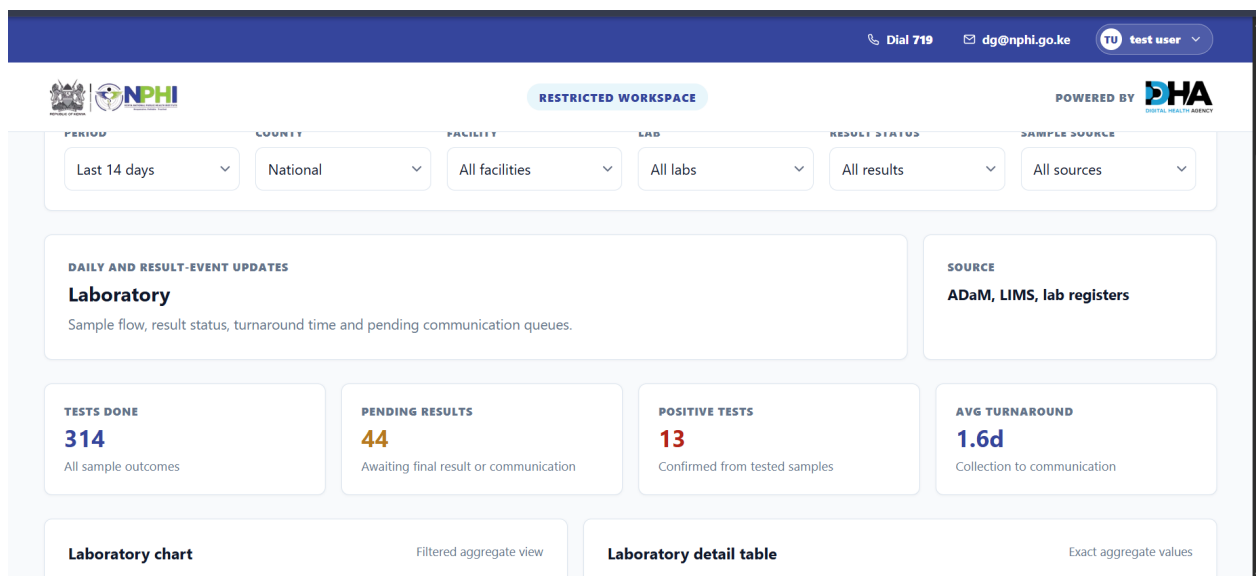


Figure A4. Laboratory KPI cards before the agreed three-indicator simplification.

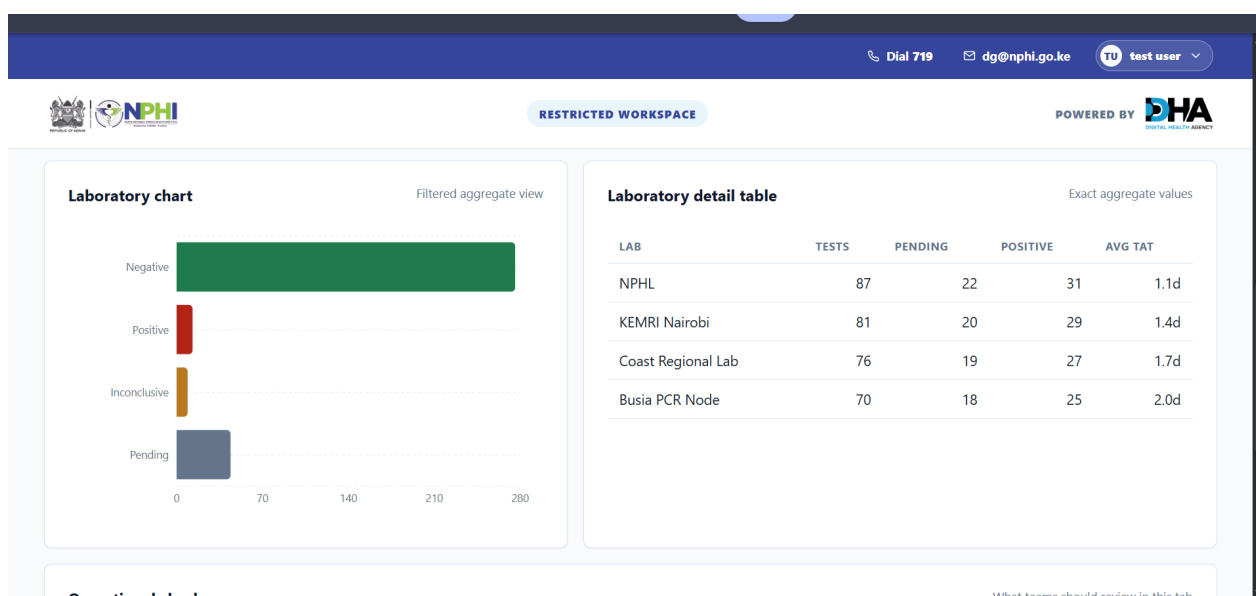


Figure A5. Laboratory chart and detail table used during the discussion on drill-down, test counts and turnaround time.

END OF IMPLEMENTATION GUIDE